CHAPTER 25 · DERIVATIONS

NUCLEAR PHYSICS

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derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 RADIOACTIVE DECAY LAW

SYLLABUS 26.2(F)

RESULT $N = N_0 e^{-\lambda t}$

1. Decay law: $\frac{dN}{dt} = -\lambda N$.
2. Separate: $\frac{dN}{N} = -\lambda\, dt$.
3. Integrate from N_0 at $t = 0$: $\ln \frac{N}{N_0} = -\lambda t$.
4. **$N = N_0 e^{-\lambda t}$.**

2 HALF-LIFE

SYLLABUS 26.2(G)

RESULT $t_{½} = \frac{\ln 2}{\lambda}$

1. At $t = t_{½}$, $N = \frac{N_0}{2}$.
2. $\frac{N_0}{2} = N_0 e^{-\lambda t_{½}}$, so $e^{\lambda t_{½}} = 2$.
3. $\lambda t_{½} = \ln 2$, so **$t_{½} = \frac{\ln 2}{\lambda}$** $\approx \frac{0.693}{\lambda}$.